



# Liquid fuels demand and supply model

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## Week 1 analytical pack

South Africa | Petrol and diesel

6 October 2026

Internal working discussion

Photo source: Royal Vopak | Terminal Durban | vopak.com



# Resolve the SA fuel outlook, then test Vopak's accessible market

## Situation → Complication → Resolution | evidence, actions and delivery roadmap

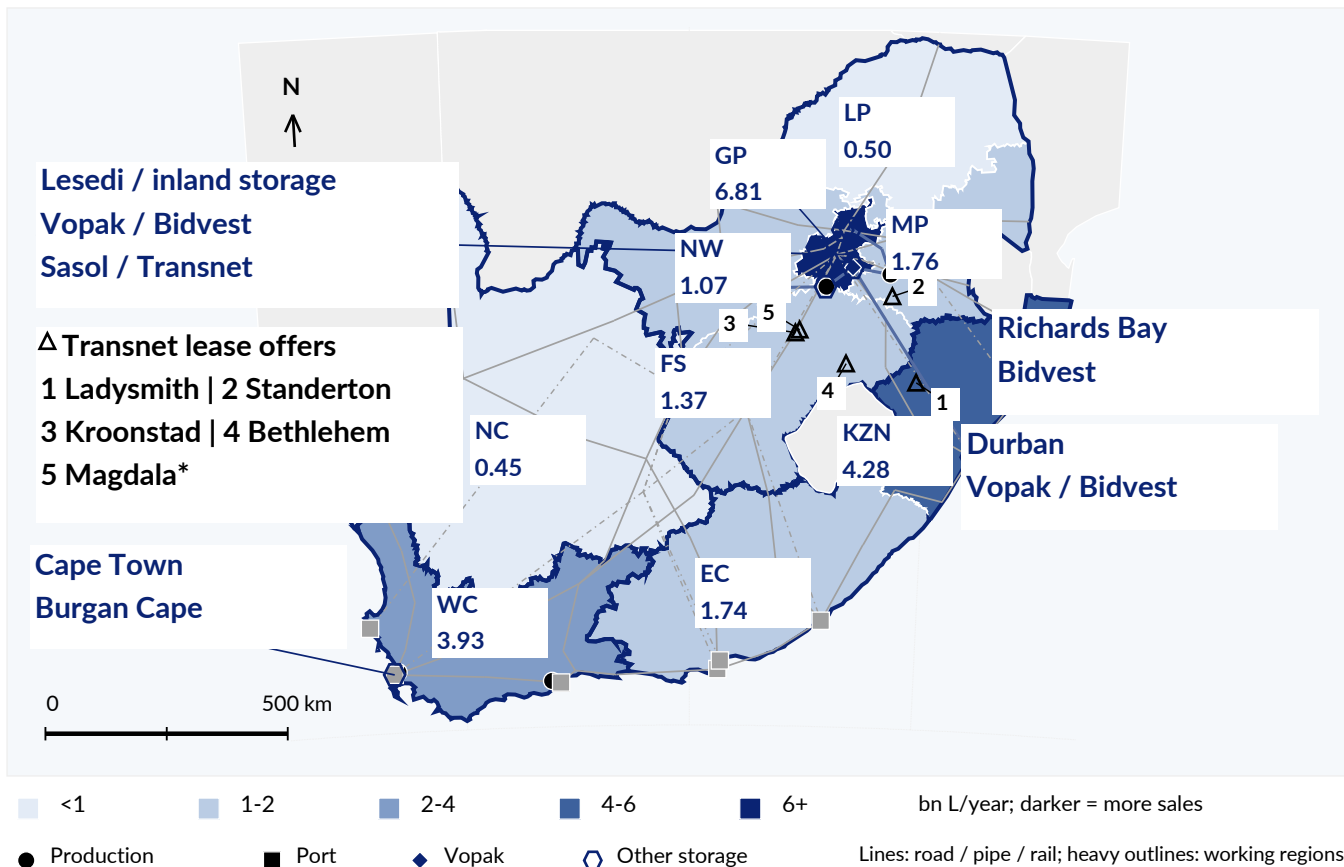
| SCR reference          | Evidence and required next step                                                                                                                                                 | Proposed lead                        | Timing and output                                     | Deep dives                   |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|-------------------------------------------------------|------------------------------|
| <b>S1 Situation</b>    | Demand baseline and regional market<br>Have: provincial sales history and map. Reconcile six flagged years; refresh a complete year.                                            | Manish<br>Nigel review               | W1–W2<br>Sourced demand baseline                      | <a href="#">3 / 4</a>        |
| <b>S2 Situation</b>    | Domestic supply, finished imports and entry ports<br>Have: 2024 sales/trade; diesel imports conflict. Gap: matched output/stocks and product-specific port allocation.          | Manish<br>Nigel access               | W1–W3<br>Fuel balance + port bridge                   | <a href="#">5 / 6</a>        |
| <b>C1 Complication</b> | Demand drivers and levers<br>Have: indexed power, road/rail, BEV/hybrid sales and real GDP. Refresh prices; calibrate baseline/alternatives into litres effects.                | Manish<br>Nigel review               | W1 define; W2–W4 test<br>Lever-to-demand bridge       | <a href="#">7</a>            |
| <b>C2 Complication</b> | Refinery and plant scenarios<br>Have: published capacity history and conditional CEF case. Agree timing/yields; resolve repeated keys and plant mechanisms.                     | Manish<br>Henry / Nigel              | W1–W3<br>Plant output + imports                       | <a href="#">8</a>            |
| <b>C3 Complication</b> | Economic accessibility and competing routes<br>Cost map is illustrative. Compare Durban–NMPP–Lesedi, road/rail, Matola and Walvis Bay for the same product and destination.     | Manish<br>Nigel / Henry              | W1 scope; W2–W4 test<br>Delivered R/litre + access    | <a href="#">9 / 10</a>       |
| <b>R1 Resolution</b>   | Actual Vopak share and contestable customer demand<br>Share is unknown. Test each candidate increment against cost, physical capacity, commercial rights and unique deliveries. | Nigel client data<br>Manish analysis | W1 request; W2–W4 test<br>Served + contestable litres | <a href="#">11 / 12 / 13</a> |
| <b>R2 Resolution</b>   | Storage and optional investment<br>Partial published stocks are available. Verify usable product capacity; size additional service needs only after the incremental flow case.  | Nigel priority<br>Manish / Henry     | W4–W6 if prioritised<br>Flow-to-stock / investment    | <a href="#">14 / 15</a>      |

Petrol + diesel; jet separate. Owners and weeks are proposed. Open gaps and illustrations remain explicitly labelled.

SCR S1 | Situation | Return to overview: page 2

## Locate fuel demand alongside the infrastructure serving it

### Petrol + diesel sales by province | 2022, bn litres/year



### Evidence and implications

#### 01 | Demand is concentrated

Gauteng, KwaZulu-Natal and Western Cape account for 68.6% of reported 2022 petrol/diesel sales.

#### 02 | Regions are working groups

Province boundaries define reporting groups; they do not establish terminal catchments.

#### 03 | Sales are historical demand evidence

The map is sourced provincial sales, rather than a current-year outlook or commercial accessibility result.

#### Next steps | proposed owners

Manish: source a later complete year and test actual destinations, route costs and customer access.

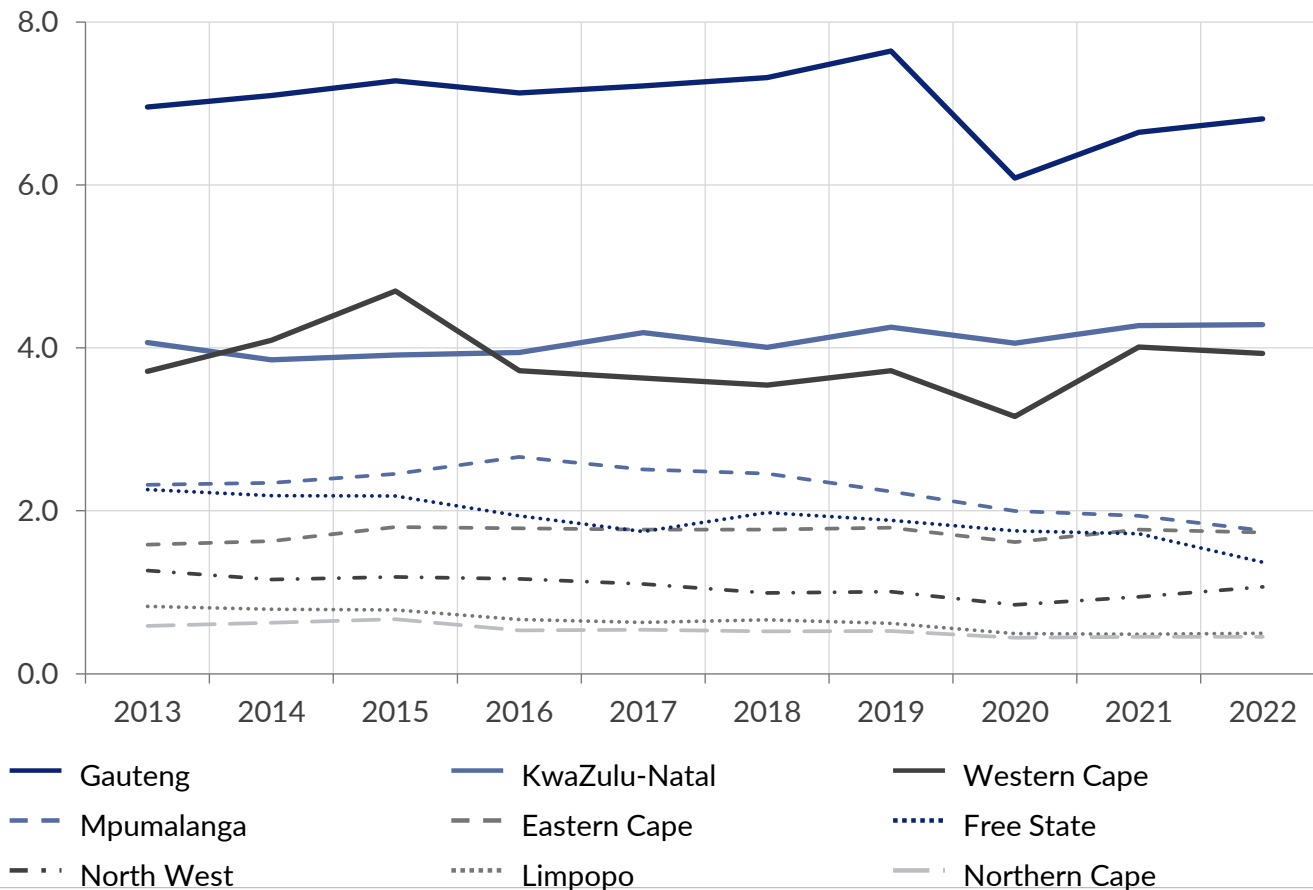
Lease offers are not operating capacity. \*Magdala RFP location narrative conflicts; briefing coordinate used. Jet excluded.

Source: DMPR 2022 sales; geoBoundaries/OCHA/MDB 2020 provinces (CC BY 3.0 IGO); operator sources on slide 12.

SCR S1 | Situation | Return to overview: page 2

## Track petrol and diesel sales by province over time

### Annual provincial sales | 2013–2022, bn litres/year



### Evidence and implications

#### 01 | History ends in 2022

The latest complete provincial year in this extract is 2022.

#### 02 | 2023 is incomplete

Only Q1 is present; it is excluded from annual comparisons.

#### 03 | Six years are flagged

Provincial sums differ from national sales in 2013, 2014, 2015, 2017, 2018 and 2021.

### Next steps | proposed owners

Manish: reconcile each flagged value to its original workbook cell and log old value, proposed value and reason.

[SCR S2 | Situation | Return to overview: page 2](#)

# Collect the national fuel balance and expose source disagreements

## National petrol/diesel | 2024, billion litres

COLLECTED | trade and sales; production/stock reconciliation remains open

| Product  | Reported sales | Imports | Exports | Net imports |
|----------|----------------|---------|---------|-------------|
| Petrol   | 9.029          | 4.000   | 0.920   | 3.080       |
| Diesel   | 11.734         | 10.800  | 0.790   | 10.010      |
| Combined | 20.763         | 14.800  | 1.710   | 13.090      |

### Source flag | 2024 diesel imports

FIASA staged: 14.793 | Government report: 10.800

Difference: -3.993 bn L. Preserve both; investigate scope and vintage.

### Domestic production remains a dated observation

Latest staged balance: 2021 petrol 6.35; diesel 5.31 bn L.

Do not combine 2021 output with 2024 sales/trade to close a balance.

## Evidence and implications

### 01 | Public figures are collectable

2024 petrol/diesel imports total 14.8 bn L in the government report. Exports total 1.71 bn L; these are rounded reported figures.

### 02 | The sources disagree

FIASA reports 14.793 bn L of diesel imports for 2024 versus 10.8 bn L in the government report. Neither value has been silently replaced.

### 03 | Port cargo has wider coverage

TNPA publishes liquid-bulk cargo by port, but it includes crude and other liquids. It does not establish petrol/diesel import allocation.

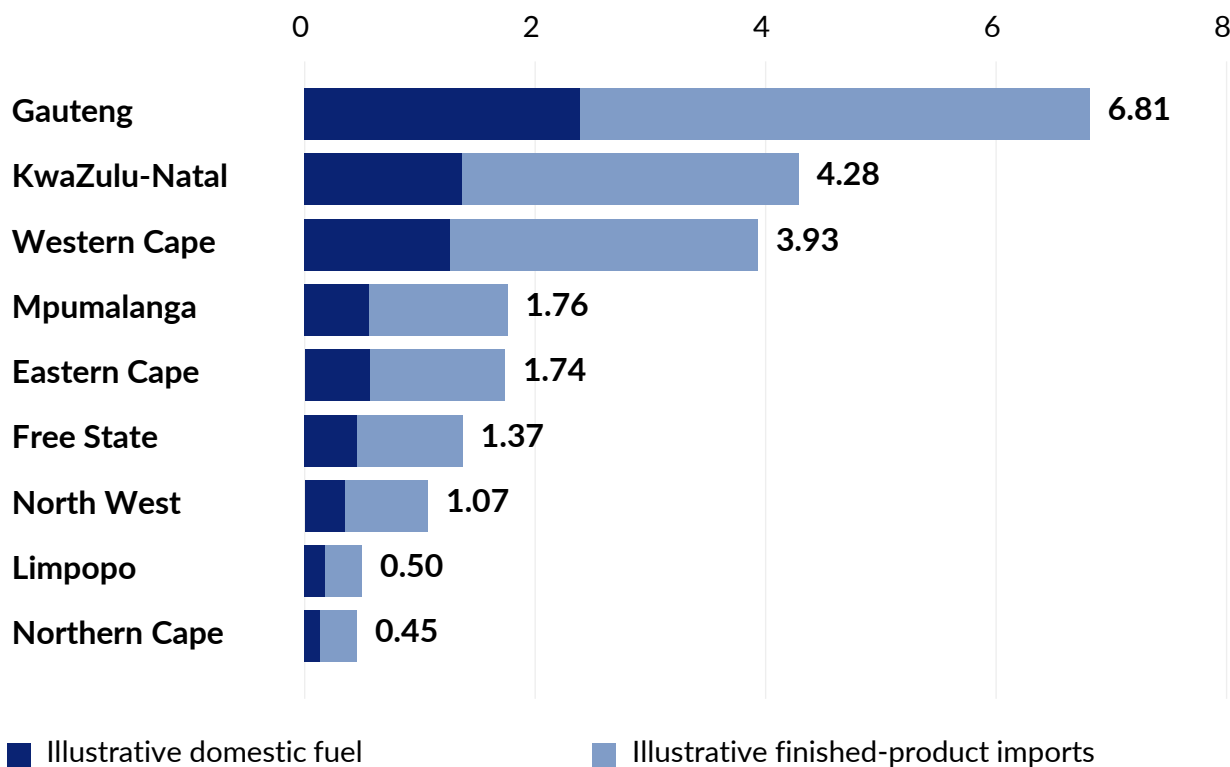
### Next steps | proposed owners

Manish: reconcile raw SARS product codes and units, then source matched-year production and stocks. Nigel: confirm access to product-specific port data.

[SCR S2 | Situation | Return to overview: page 2](#)

# Shape the provincial split between domestic fuel and imports

## 2022 sales totals | ILLUSTRATIVE domestic/import allocation



Units: bn litres/year. Imported/domestic segments are a scenario illustration, not measured provincial flows.

## Evidence and implications

### 01 | Demand totals are sourced

The provincial bars use reported 2022 petrol/diesel sales.

### 02 | Origin shares are illustrative

Domestic/import fractions are authored examples; no actual provincial origin allocation is established.

### 03 | Imports need a national balance

Exports and stock movements must be reconciled before a sourced import requirement can be presented.

## Next steps | proposed owners

Manish: populate the matched-year national balance first; retain provincial origin as unknown until supported.

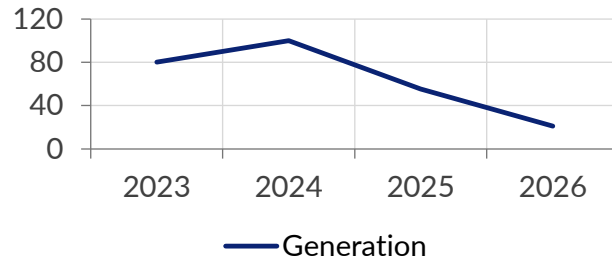
SCR C1 | Complication | Return to overview: page 2

# Compare the observed drivers of petrol and diesel demand

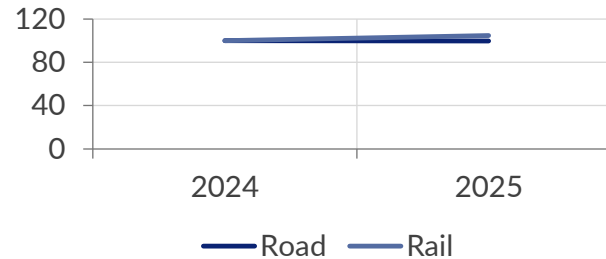
## Demand-driver dashboard | every series indexed to 2024 = 100

Observed series; common 0–120 scale. FY ends March; CY is calendar year.

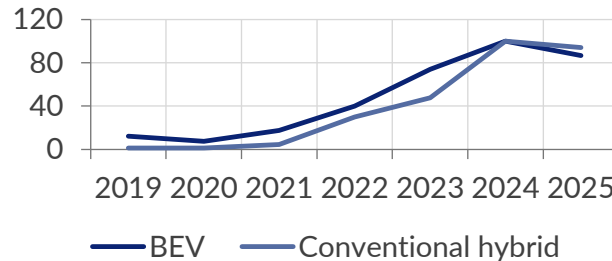
### Power | Eskom + IPP OCGT, FY



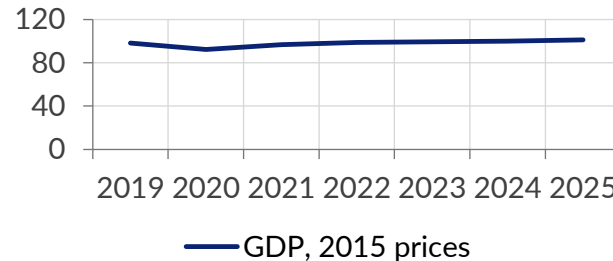
### Freight | all goods payload, CY



### Fleet uptake | annual new sales, CY



### Activity | real GDP, CY



Price flag: annual inland petrol-95 ends in 2023 at R23.14/L (nominal); not rebased or extrapolated.

## Evidence and implications

### 01 | Power generation has fallen

OCGT generation falls from 5,143 GWh in FY2024 to 1,079 in FY2026 (-79%). This is generation, not diesel litres.

### 02 | Freight needs a consistent vintage

2025 versus 2024: rail +4.7%; road -0.3%. Road 2024 was revised from 790.6 to 979.8 million tonnes; no old/new splice.

### 03 | Sales are not fleet penetration

2025 sales: 1,088 BEVs and 12,818 conventional hybrids. Hybrids still consume fuel; GDP movement alone does not establish kilometres.

## Next steps | proposed owners

Manish: translate generation, freight tonne-km, fleet cohorts and real price/activity into sourced litres effects. Nigel: agree alternatives. Refresh prices; calibrate before forecasting.

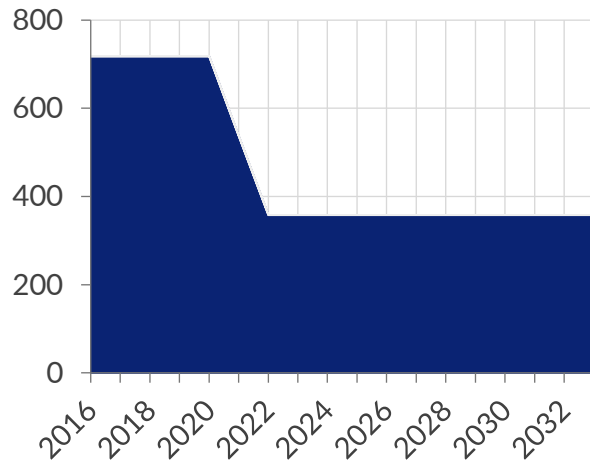
SCR C2 | Complication | Return to overview: page 2

## Compare the capacity history with a conditional redevelopment case

### Published footprint + illustrative scenarios | thousand bbl/day

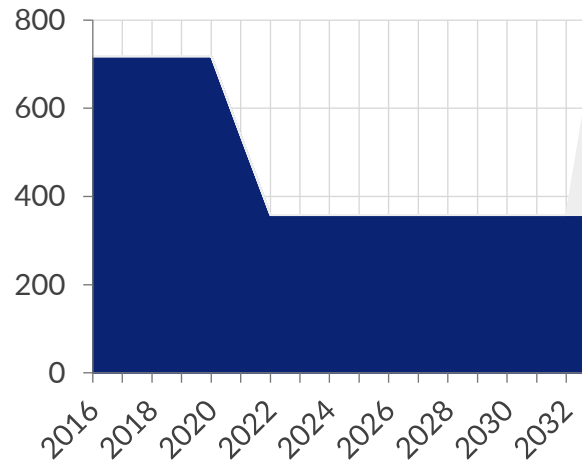
History 2016–2025 is sourced; 2026 onward is illustrative, not a forecast

#### A | No new refinery capacity



2033: 358 thousand bbl/day

#### B | Conditional CEF redevelopment



2033: 758 thousand bbl/day

718 → 538 → 358: published footprint; idle nameplate and synthetic crude-equivalent included.

B assumes illustrative FID 2029 + 48 months; no sanctioned date. Both hold existing capacity flat.

### Evidence and implications

#### 01 | History shows a structural decline

Published total falls from 718 in 2016–2020 to 538 in 2021 and 358 from 2022–2025. Capacity does not measure realised output.

#### 02 | Redevelopment is conditional

CEF proposes approximately 400 thousand bbl/day after FID. Construction is approximately 48 months; the 2029 FID here is authored for comparison.

#### 03 | Product output still needs modelling

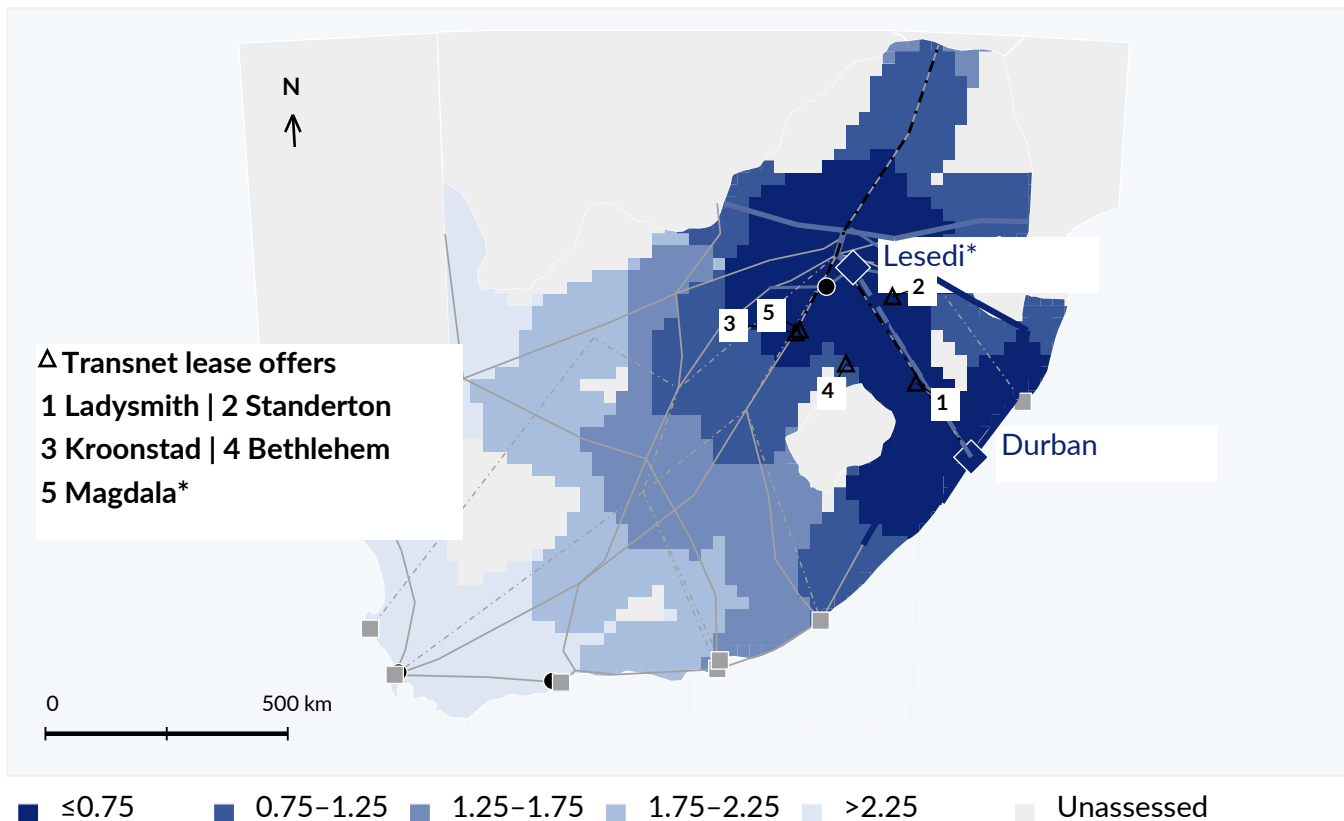
Source utilisation, downtime and product yields before translating capacity into domestic supply. Secunda/Natref cases remain open.

### Next steps | proposed owners

Nigel: agree scenario timing. Manish: reconcile plant capacity and output/yields. Henry: review technical feasibility and the Sasol gas-to-liquids mechanism.

# Map conditional market access from Durban and Lesedi

## Road-delivery accessibility | illustrative transport cost, R/litre



Example rate: R0.15/L dispatch + R0.002/L/km. Grey = unassessed.

Illustrative road cost only. Pipeline/rail access and commercial rights are not priced.

Source: Existing appendix routes; illustrative candidate connections, not verified catchments. Natural Earth 1:50m.

## Evidence and implications

### 01 | Darker cells mean lower example cost

The surface shows illustrative delivered road-transport cost in R/litre.

### 02 | Other modes remain context

Pipeline and rail costs, capacity and access are not evaluated in this surface.

### 03 | Physical proximity is insufficient

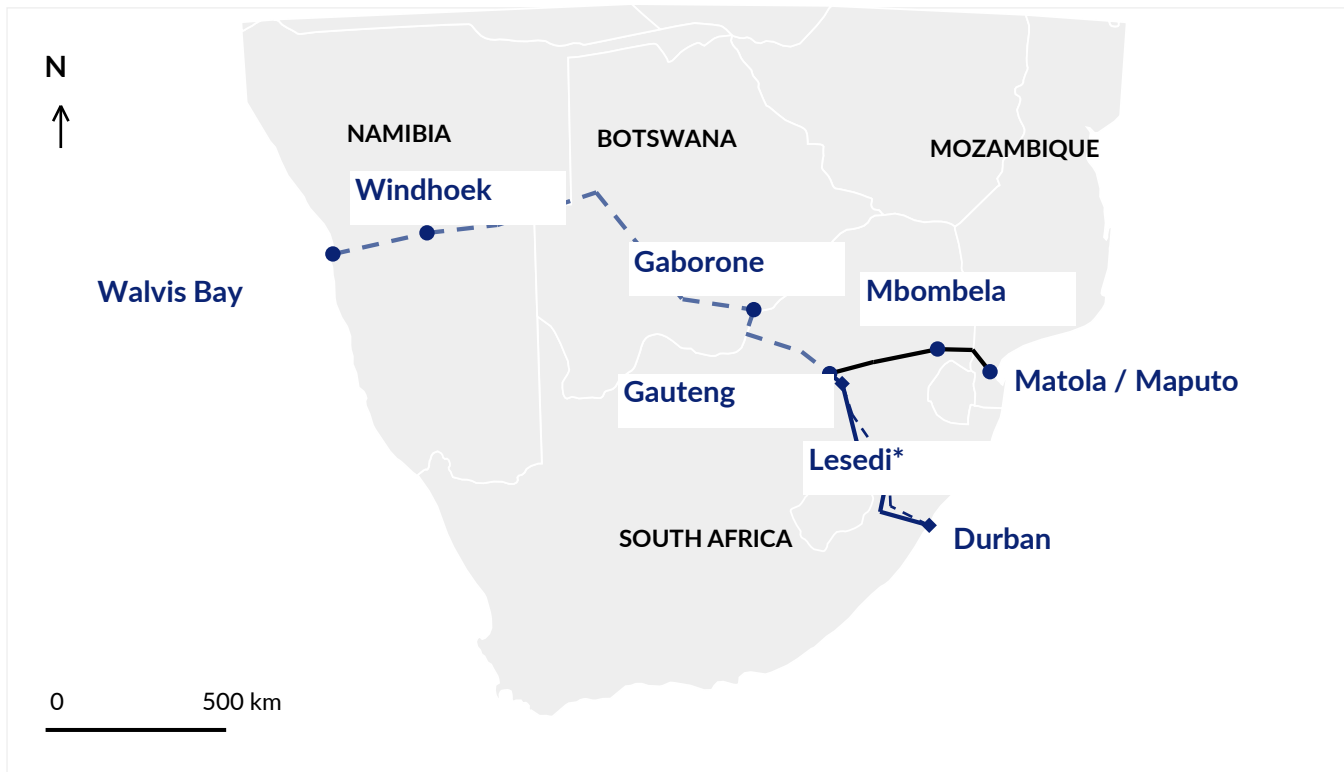
No operational or commercial catchment is validated by this illustration.

### Next steps | proposed owners

Manish and Nigel: select the first destination and obtain dated route tariffs, quotations and access evidence.

# Locate alternative gateways to the same inland customers

## Durban, Matola and Walvis Bay | schematic corridor comparison



Solid: Durban / N4 candidates | Dashed: Trans-Kalahari / NMPP

LAEA / WGS84; routes schematic. No cost, available capacity, fuel rights or catchment is inferred.

## Evidence and implications

### 01 | Compare the same inland destination

Gauteng is the common comparison market. Lesedi is a Vopak facility, not a destination every competing route must pass through.

### 02 | Matola approaches from the east

N4/Maputo connects Mozambique to Gauteng via Komatipoort and Mpumalanga. Fuel-compatible terminal access and road/rail service need testing.

### 03 | Walvis is an inland comparator

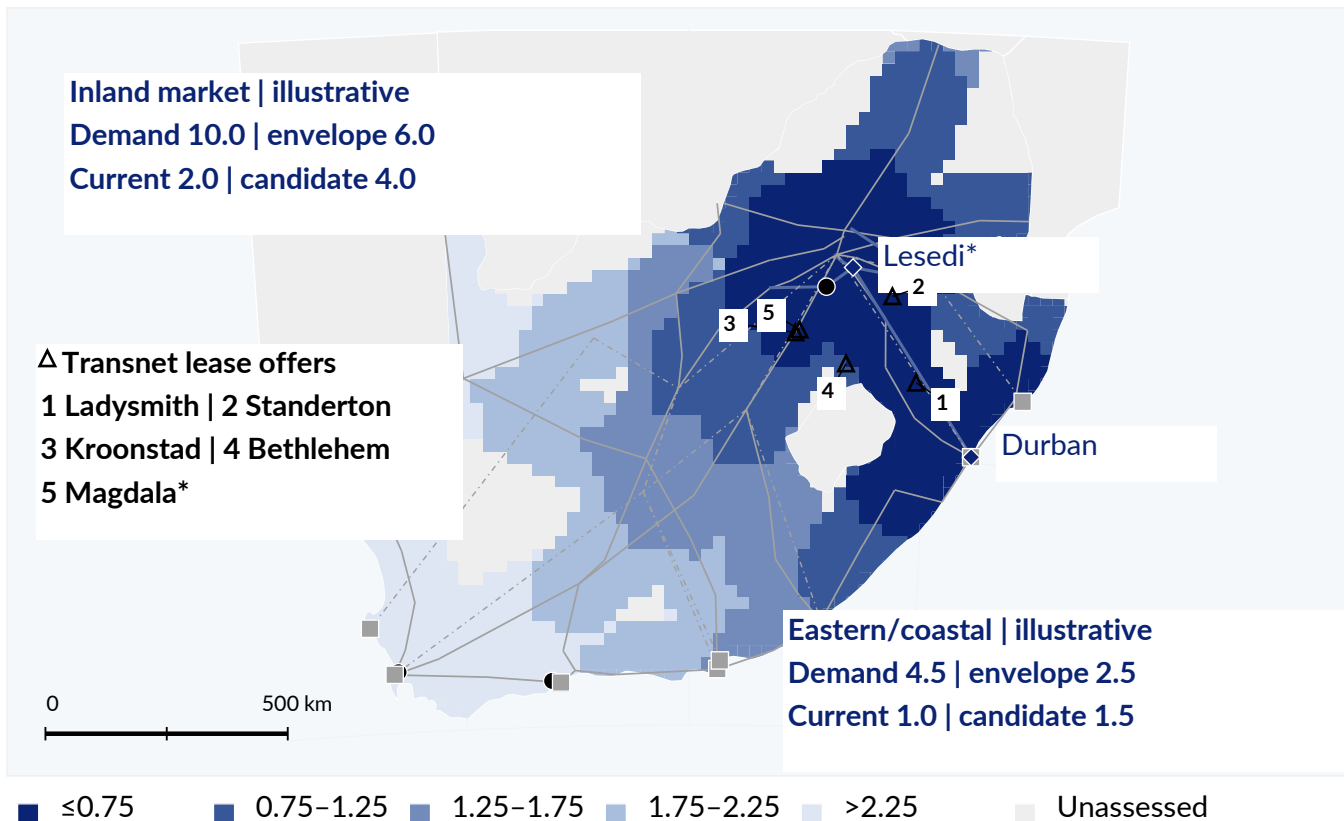
Trans-Kalahari links Walvis Bay via Namibia and Botswana to Gauteng. Its competitiveness for specific inland customers is unassessed.

## Next steps | proposed owners

Nigel: select the customer destination. Manish: compare handling, transport, border/FX costs and access on the same R/litre basis; keep unevidenced routes unassessed.

# Separate demand, accessible volume and Vopak flows

## Illustrative road cost (R/litre) and market volumes



**Receipts 2.8 + 2.0 – transfer 1.8 = unique demand 3.0**

Example rates: R0.15/L + R0.002/L/km. Grey = unassessed. Volumes: bn L/year.

Source: Illustrative CSVs and road transport rates; no actual Vopak throughput or quoted transport costs supplied.

## Evidence and implications

### 01 | Balance arithmetic is illustrative

Example demand 21.0 less domestic production 7.0 gives imports 14.0, before exports and stocks.

### 02 | Transfers count once

Example Durban receipts 2.8 plus Lesedi receipts 2.0 less shared transfer 1.8 gives unique served demand 3.0.

### 03 | Candidate volume is conditional

The example envelope is 8.5 and additional candidate volume 5.5; neither is forecast capture.

## Next steps | proposed owners

Nigel: request client flows. Manish: replace examples with matched-year product and destination records.

SCR R1 | Resolution | Return to overview: page 2

## Size each regional market and locate competing storage

### All four regions | reported 2022 demand, bn litres/year

National total 21.90 bn L | petrol + diesel; jet excluded

| Region / provinces       | Reported demand | Vopak served    | Additional contestable |
|--------------------------|-----------------|-----------------|------------------------|
|                          | 04812           |                 |                        |
| Eastern coast<br>EC/KZN  | 6.02            | Not established | Not established        |
| Inland<br>GP/FS/LP/MP/NW | 11.50           | Not established | Not established        |
| Western coast<br>WC      | 3.93            | Not established | Not established        |
| Other<br>Northern Cape   | 0.45            | Not established | Not established        |

To calculate share: unique Vopak customer deliveries / same-year regional demand.

Contestable demand needs route cost, compatible tanks, contracts and switching evidence.  
Count Durban-Lesedi transfers once. Region boundaries are working groupings, not catchments.

[1] Vopak [2] Bidvest [3] Burgan [4] Shell [5] NERSA [6] Sasol [7] Transnet

Source: DMPR 2022 provincial petrol/diesel sales; public operator sources [1-7], checked 6 Oct 2026. Share not established.

### Regional competitor evidence

#### 01 Eastern coast

Vopak: Durban [1]. Bidvest: Durban and Richards Bay [2]; mixed-product capacity.  
Transnet: Ladysmith lease tanks [7].

#### 02 Inland

Vopak: Lesedi [1]. Bidvest: Isando [2].  
Sasol: Alrode, Pretoria West, Waltloo, Sasolburg [6].  
Transnet: Tarlton / Jameson Park + four lease sites [7].

#### 03 Western coast

Burgan Cape: Cape Town petrol/diesel terminal [3].  
Confirm current tank availability and customer access.

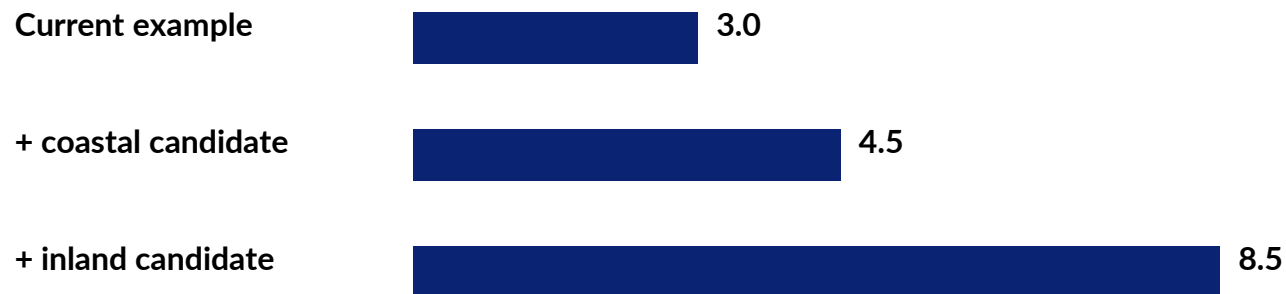
#### 04 Other / Northern Cape

Operator coverage incomplete. Check Shell site notices [4] and NERSA licences/access records [5].

# Show what must hold to reach each candidate volume

## Accessibility gates and volume milestones | illustrative bn L/year

ILLUSTRATIVE | each increment requires evidence; not cumulative terminal receipts



## What must be true for the additional volume?

| Gate              | Coastal +1.5                  | Inland +4.0                           |
|-------------------|-------------------------------|---------------------------------------|
| Delivered cost    | ≤ agreed customer threshold   | Competitive full route R/litre        |
| Physical access   | Compatible receipt / dispatch | NMPP or alternative route capacity    |
| Commercial rights | Customer access / switching   | Destination / contract access         |
| Volume accounting | Unique final deliveries       | Remove shared Durban–Lesedi transfers |

## Evidence and implications

### 01 | Reach is necessary, not sufficient

Accessibility map p9 and competing routes p10 locate candidate access. Neither currently links a verified cost threshold to customer litres.

### 02 | Example milestones are conditional

The current example is 3.0. Coastal candidate 1.5 raises it to 4.5; inland candidate 4.0 raises it to 8.5. These are authored volumes.

### 03 | Every increment needs four gates

Evidence cost, route/tank capacity, customer rights and unique deliveries for the same product and period. Unassessed litres remain explicit.

## Next steps | proposed owners

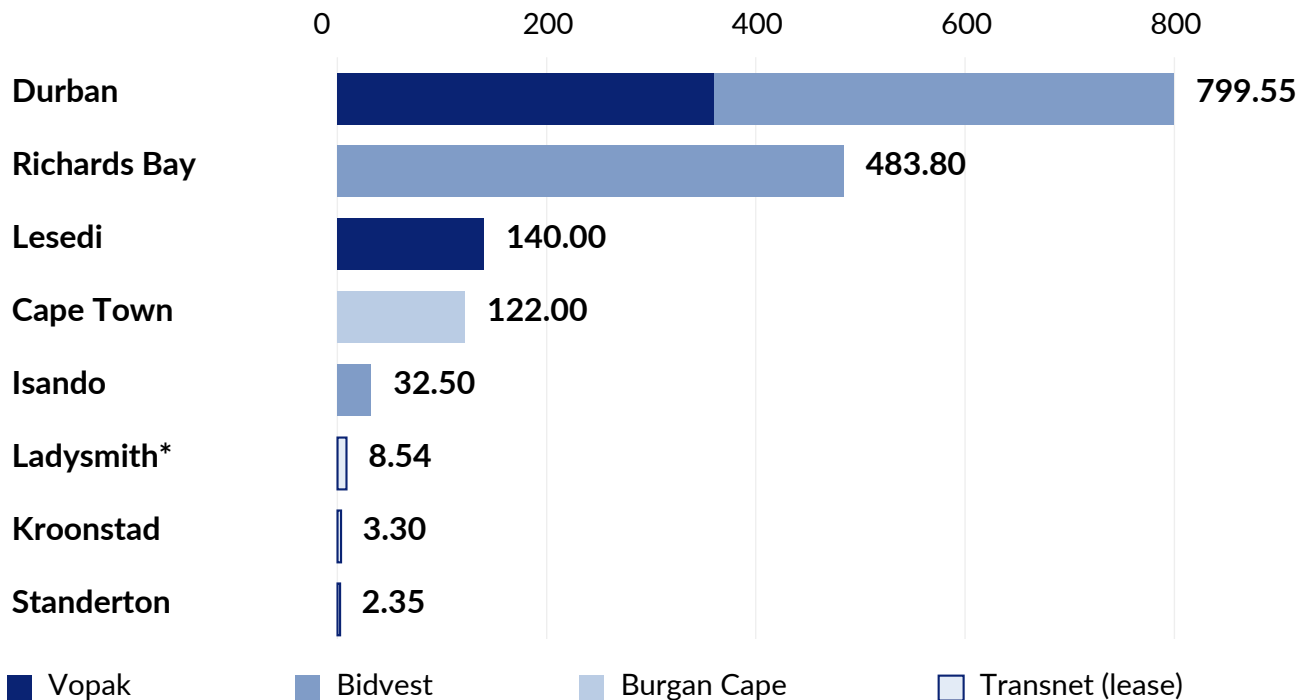
Nigel: agree destinations and acceptable delivered cost. Manish: build a customer-by-route gate table; record passed, failed and unassessed litres, with sources and conditions.

[SCR R2 | Resolution | Return to overview: page 2](#)

# Compare published storage capacity by location

## Published storage by location | thousand m<sup>3</sup>; mixed capacity bases

One shared scale for every location



Small lease bars reflect smaller tank volumes; they are not operating supply.

\*Ladysmith excludes 0.474 intermixture. Unquantified: Bethlehem, Magdala, Sasol, Tarlton, Jameson Park.

Source: Vopak; Bidvest; VTTI Burgan Cape; Transnet lease RFPs (5 Oct 2026), pp. 2-3. Checked 6 Oct 2026.

## Evidence and implications

### 01 | Capacity is a stock

Published tank volumes are not annual fuel deliveries or market share.

### 02 | The inventory is partial

Missing capacities are unknown, rather than zero; mixed product scopes limit comparison.

### 03 | Leases are conditional

Transnet lease tanks are not counted as currently available operating supply.

## Next steps | proposed owners

Manish: verify product compatibility, capacity basis and operating status. Nigel: confirm usable client capacity.

# Gate investment on evidenced incremental customer flows

## Resolution | optional service and investment assessment

EVIDENCE GAP | analysis specification, not a populated result

### New storage and investment scope

Incremental accessible flows, turnover/peak inventory, product compatibility, cost and commercial return criteria.

Output: Throughput-to-usable-stock bridge with seasonality, inventory days and dispatch limits; downside sensitivity; no investment claim from import deficit alone.

### Evidence required before quantification

Partial gross/lease stocks, without client handling/inventory profile.

Detailed service sizing deferred; infrastructure secondary to fuel integration.

## Evidence and implications

### 01 | Evidence status

These topics remain open. Existing calculations or public source leads do not establish calibration or closure.

### 02 | What is already available

Partial gross/lease stocks, without client handling/inventory profile.

### 03 | What this page will resolve

A traceable petrol/diesel result with source, period, units and explicit unassessed residual. Jet stays separate.

### Next steps | proposed owners

Nigel: decide whether to prioritise investment analysis after the flow case. Manish and Henry: bridge incremental flows to usable stock and service constraints.

# Agree priorities and next steps

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Confirm demand, routes and access

6 October 2026

Internal working discussion

Photo source: Royal Vopak | Terminal Lesedi | vopak.com

